

Chapter 04: Vector space:

1) Definition of vector space:

Let F be field of real numbers $(\mathbb{R}, +, \cdot)$ or field of complex numbers $(\mathbb{C}, +, \cdot)$ and let the following operators be defined for:

$$(+): V \times V \rightarrow V$$

$$(u, v) \rightarrow u + v \text{ (Internal operator)}$$

$$(\cdot): F \times V \rightarrow V$$

$$(\lambda, u) \rightarrow \lambda u \text{ (External operator)}$$

where V is set of vectors.

V is vector space over F if:

$$1/ \exists 0_v \in V \mid \forall u \in V, u + 0_v = u \text{ (identity element)}$$

$$2/ \forall u \in V, \exists -u \in V \mid u + (-u) = 0_v \text{ (inverse)}$$

$$3/ \forall u, v, w \in V, u + (v + w) = (u + v) + w \text{ (associativity)}$$

$$4/ \forall u, v \in V, u + v = v + u \text{ (commutativity)}$$

$$5/ \exists 1 \in F \mid \forall u \in V, 1 \cdot u = u \text{ (identity element of } F)$$

$$6/ \forall \alpha, \beta \in F, \forall u \in V, (\alpha \cdot \beta) \cdot u = \alpha \cdot (\beta \cdot u) \text{ (associativity)}$$

$$7/ \forall u \in V, \forall \alpha, \beta \in F, (\alpha + \beta) \cdot u = \alpha \cdot u + \beta \cdot u$$

$$8/ \forall u, v \in V, \forall \lambda \in F, \lambda \cdot (u + v) = \lambda \cdot u + \lambda \cdot v \text{ (distributivity law)}$$

2) Examples of vector spaces:

$$1) \mathbb{R}^n = \{(x_0, x_1, \dots, x_n) \mid x_k \in \mathbb{R}, k \in \mathbb{N}\} \text{ like } (5, 3, -\pi) \in \mathbb{R}^3$$

$\mathbb{C}^n$ also, and the identity element is $(0, 0, \dots, 0)$ (coordinate space)

$$2) \mathbb{R}[X] = \{P \mid P(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \dots \text{ where } \alpha_k \in \mathbb{R}, k \in \mathbb{N}\}$$

like $P \in \mathbb{R}[X]$ where $P(x) = 9x^3 - 5x + 6$ same with $\mathbb{C}[X]$

$$*) \mathbb{R}_n[X] = \{P \mid \deg P \leq n\} \text{ like for } P(x) = x^3 - 1 + x \Rightarrow \deg P = 3$$

$$\text{and } G(x) = -x \Rightarrow \deg P = 1$$

$$\text{and } H(x) = \pi \Rightarrow \deg H = 0$$

$$\text{and } T(x) = 0 \Rightarrow \deg = -\infty \text{ (special case)}$$

and all P, G, H, T belong to $\mathbb{R}_3[X]$

and the identity element is $0_{\mathbb{R}[X]}$ where $0_{\mathbb{R}[X]}(x) = 0$ and $\deg(0_{\mathbb{R}[X]}) = -\infty$
(polynomial space)

$$3) \mathcal{F}(D, C) = \{f \mid f: D \xrightarrow{x \mapsto f(x)} C\} \text{ like } f \in \mathcal{F}(\mathbb{R}_+^*, \mathbb{R}) \text{ where } f(x) = \ln x$$

(function space)

$$4) M_{n \times m}(\mathbb{R}) \text{ (Matrix space)}$$

3) Properties of vector space:

$$0 \cdot u = 0_v$$

$$\alpha \cdot 0_v = 0_v$$

$$(-\alpha)u = \alpha(-u) = -\alpha u$$

$$(-\alpha)(-u) = \alpha u$$

$$\alpha u = 0_v \Rightarrow \alpha = 0 \vee u = 0_v$$

4) Vector subspace:

if $U \subset V$, U is vector subspace of V if:

1) $0_v \in U$ (identity element)

2) $\forall u, v \in U, u+v \in U$ (closed under addition)

3) $\forall \alpha \in F, u \in U, \alpha \cdot u \in U$ (closed under multiplication)

$$\textcircled{2} \wedge \textcircled{3} \Leftrightarrow \forall \alpha \in F, \forall u, v \in U, \alpha \cdot (u+v) \text{ or } \alpha \cdot u + \alpha \cdot v \in U$$

$$\text{Ex: } U = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(-x) = f(x)\}$$

1) for $f = 0_{\mathcal{F}(\mathbb{R}, \mathbb{R})}$, $f(x) = 0$ for any x

$$\text{so } f(-x) = 0 \text{ therefore } f(-x) = f(x) \Rightarrow 0_{\mathcal{F}(\mathbb{R}, \mathbb{R})} \in U$$

2) $\forall f, g \in U$ we have: $\begin{cases} f(-x) = f(x) \\ g(-x) = g(x) \end{cases}$

$$(f+g)(-x) = f(-x) + g(-x) = f(x) + g(x) = (f+g)(x)$$

$$\text{so } (f+g)(-x) = (f+g)(x) \Rightarrow f+g \in U$$

3) $\forall \alpha \in \mathbb{R}, \forall f \in U$ so $f(-x) = f(x)$

$$\alpha f(-x) = \alpha f(x)$$

$$(\alpha f)(-x) = (\alpha f)(x) \Rightarrow \alpha f \in U$$

Therefore U is vector subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$

note: vector subspace is also a vector space.

Note:

$$(5, 3, -1) + (0, 3, 1) = (5+0, 3+3, -1+1)$$

$$(5, 3, -1) + (0, 3, 1) = (5, 6, 0)$$

for computing coordinate vectors

$$3(5, 3, -1) = (15, 9, -3)$$

7) Intersection of vector subspaces:

E_1 and E_2 are vector subspaces of V so $E_1 \cap E_2$ is also a vector subspace of V .

pf: 1) $0_V \in E_1 \cap E_2$ because E_1 and E_2 are vector subspaces.
so $0_V \in E_1 \cap E_2$

$$\begin{aligned} 2) \forall \alpha \in F, \forall u, v \in E_1 \cap E_2 &\Rightarrow u \in E_1 \wedge u, v \in E_2 \\ \alpha \cdot (u+v) \in E_1 &\Rightarrow \alpha \cdot (u+v) \in E_2 \\ \Rightarrow \alpha \cdot (u+v) &\in E_1 \cap E_2 \end{aligned}$$

Ex:

$$E_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x+y=3z\} \text{ is vector subspace of } \mathbb{R}^3$$

$$E_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x-y=z\} \text{ is vector subspace of } \mathbb{R}^3$$

$$\text{so } E_1 \cap E_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x+y=3z \wedge x-y=z\}$$

$$\begin{cases} x+y=3z \\ x-y=z \end{cases} \Rightarrow \begin{cases} 2x=4z \\ 2y=2z \end{cases} \Rightarrow \begin{cases} x=2z \\ y=z \end{cases}$$

$$\text{so } E_1 \cap E_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x=2z \wedge y=z\}$$

$$E_1 \cap E_2 = \{(2z, z, z) \in \mathbb{R}^3\}$$

where $E_1 + E_2$ is also vector subspace of $\mathbb{R}^3$

Warning:

$E_1 \cup E_2$ is not always a vector subspace (it has no rule for union)

Spanning set, linear combination, dimension, bases, linear independence:

1) Linear combination:

if u is linear combination of $v_1, v_2, \dots, v_n$ then:

$$u = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n \text{ where } \alpha_k \in \mathbb{R}$$

Ex:

for example prove $(-1, -3, 1)$ is linear combination $(1, 0, 2), (0, 1, -1)$:

$$(-1, -3, 1) = \alpha(1, 0, 2) + \beta(0, 1, -1)$$

$$\begin{cases} \alpha \neq 0 \neq 1 \\ 0 \neq \beta = -3 \\ 2\alpha - \beta = 1 \end{cases} \Rightarrow \begin{cases} \alpha = -1 \\ \beta = -3 \\ 2(-1) - (-3) = 1 \checkmark \end{cases}$$

so $(-1, -3, 1)$ is linearly combination of $(1, 0, 2), (0, 1, -1)$.

2) Linear independence:

if for $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \dots + \alpha_n v_n = 0_v \Rightarrow \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$
Then $v_1, v_2, v_3, \dots, v_n$ is linearly independent.

Ex:

prove $(0, 1, 0), (2, 1, -1), (2, 3, -1)$ is linearly dependent:

$$\alpha(0, 1, 0) + \beta(2, 1, -1) + \gamma(2, 3, -1) = (0, 0, 0)$$

$$\begin{cases} 0 + 2\beta + 2\gamma = 0 \\ \alpha + \beta + 3\gamma = 0 \\ 0 \neq \beta - \gamma = 0 \end{cases} \Rightarrow \begin{cases} 2\beta + 2\gamma = 0 \\ \alpha = -\beta - 3\gamma \\ 2\beta + 2\gamma = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \beta = -\gamma \\ \alpha = -\beta - 3\gamma \end{cases} \Rightarrow \begin{cases} \alpha = -(-\gamma) - 3\gamma \\ \beta = -\gamma \end{cases} \Rightarrow \begin{cases} \alpha = -2\gamma \\ \beta = -\gamma \end{cases}$$

and γ is free so there is infinite number of solutions for α, β, γ so
 $(0, 1, 0), (2, 1, -1), (2, 3, -1)$ is not linearly independent
so it's linearly dependent.

3) Span set:

Span of $v_1, v_2, \dots, v_n$ is defined for:

$$\text{span}(v_1, v_2, v_3, \dots, v_n) = \{ \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n \mid \alpha_k \in \mathbb{R} \}$$

special case:

$$\text{when } \text{span}(v_1, v_2, \dots, v_n) = V$$

we say $v_1, v_2, \dots, v_n$ spans the vector space V .

4) Basis:

$v_1, v_2, v_3, \dots, v_n$ is the basis set for V if:

$$\text{span}(v_1, v_2, v_3, \dots, v_n) = V$$

$v_1, v_2, v_3, \dots, v_n$ is linearly independent.

Ex1:

show that $u_1 = (3, -1, 5), u_2 = (0, 0, -1), u_3 = (1, 0, 1)$ is basis for $\mathbb{R}^3$:

① $\forall u \in \mathbb{R}^3 \mid u = (x, y, z)$, we have for:

$$(x, y, z) = \alpha(3, -1, 5) + \beta(0, 0, -1) + \gamma(1, 0, 1)$$

$$\begin{cases} x = 3\alpha + 0 + \gamma \\ y = -\alpha + 0 + 0 \\ z = 5\alpha - \beta + \gamma \end{cases} \Rightarrow \begin{cases} x = 3\alpha + \gamma \\ y = -\alpha \\ z = 5\alpha - \beta + \gamma \end{cases}$$

$$\Rightarrow \begin{cases} \alpha = -y \\ \gamma = x - 3\alpha \\ \beta = 5\alpha - z + \gamma \end{cases} \Rightarrow \begin{cases} \alpha = -y \\ \beta = 5(-y) - z + \gamma \\ \gamma = x - 3(-y) \end{cases} \Rightarrow \begin{cases} \alpha = -y \\ \beta = -5y - z + \gamma \\ \gamma = x + 3y \end{cases}$$

$$\Rightarrow \begin{cases} \alpha = -y \\ \beta = -5y - z + (x + 3y) \\ \gamma = x + 3y \end{cases} \Rightarrow \begin{cases} \alpha = -y \\ \beta = x - 2y - z \\ \gamma = x + 3y \end{cases} \quad \text{so it's always possible}$$

② we have for: $\alpha(3, -1, 5) + \beta(0, 0, -1) + \gamma(1, 0, 1) = (0, 0, 0)$

$$\begin{cases} 3\alpha + \gamma = 0 \\ -\alpha + 0 = 0 \\ 5\alpha - \beta + \gamma = 0 \end{cases} \Rightarrow \begin{cases} \alpha = 0 \\ 3(0) + \gamma = 0 \\ 5(0) - \beta + \gamma = 0 \end{cases} \Rightarrow \begin{cases} \alpha = 0 \\ -\beta + 0 = 0 \\ \gamma = 0 \end{cases} \Rightarrow \alpha = \beta = \gamma = 0$$

$\Rightarrow u_1, u_2, u_3$ is linearly independent

① ∧ ② $\Rightarrow u_1, u_2, u_3$ is basis of $\mathbb{R}^3$.

Example 2:

$$E = \{(x, y, z) \in \mathbb{R}^3 \mid x - y + 4z = 0\}$$

find basis for E :

$$x - y + 4z = 0$$

$$x = y - 4z$$

$$\Rightarrow \text{for } (x, y, z) \in E \Rightarrow (x, y, z) = (y - 4z, y, z)$$

$$(x, y, z) = (y, y, 0) + (-4z, 0, z)$$

$$(x, y, z) = y(1, 1, 0) + z(-4, 0, 1)$$

$$\Rightarrow \text{span}((1, 1, 0), (-4, 0, 1)) = E - \textcircled{1}$$

and for $\alpha(1, 1, 0) + \beta(-4, 0, 1) = (0, 0, 0)$

$$\begin{cases} \alpha - 4\beta = 0 \\ \alpha + 0 = 0 \\ 0 + \beta = 0 \end{cases} \Rightarrow \alpha = \beta = 0$$

so $(1, 1, 0), (-4, 0, 1)$ is linearly independent - $\textcircled{2}$

① ∧ ② $\Leftrightarrow (1, 1, 0), (-4, 0, 1)$ is basis for E .

Definition of finite dimensional:

V is called finite-dimensional vector space if there is $v_1, v_2, \dots, v_n$ such that $\text{span}(v_1, v_2, v_3, \dots, v_n) = V$.

for example $\mathbb{R}^3$ is finite-dimensional because there is $(0, 0, 1), (0, 1, 0), (1, 0, 0)$ such that $\text{span}((0, 0, 1), (0, 1, 0), (1, 0, 0)) = \mathbb{R}^3$

* for example non-example $\mathbb{R}[X]$ there is no list of polynomials that spans $\mathbb{R}[X]$ so $\mathbb{R}[X]$ is infinite dimensional vector space.

Theorem:

Bases of same vector space V have same number of vectors.

* for example all bases of $\mathbb{R}^3$ have 3 vectors.

5) Dimension:

Dimension of V is the number of vectors of basis of V :

if $v_1, v_2, \dots, v_n$ is basis of $V \Rightarrow \dim V = n$

* for the previous example:

$$E = \{(x, y, z) \in \mathbb{R}^3 \mid x - y + 4z = 0\}$$

and $(1, 1, 0), (-4, 0, 1)$ is basis for E (2 vectors in the basis)

so: $\dim E = 2$

Vector subspaces sums and complementary vector subspaces:

1) Sum of vector subspaces: note: $E_1 + E_2 = \{u + v \mid u \in E_1, v \in E_2\}$

if E_1 and E_2 are vector subspaces then $E_1 + E_2$ is also vector subspace

for example: $E_1 = \{(x, 0, z) \mid x + z = 0\}$ is vector subspace of $\mathbb{R}^3$

$E_2 = \{(x, y, z) \mid x + z = y\}$ is vector subspace of $\mathbb{R}^3$

for $u \in E_1 \mid u = (x, 0, z)$ where $x + z = 0$

$v \in E_2 \mid v = (w, r, t)$ where $w + t = r$

we have: $u + v = (x + w, r, z + t)$

we have $\begin{cases} x + z = 0 \\ w + t = r \end{cases} \Rightarrow \begin{cases} z = -x \\ t = r - w \end{cases} \Rightarrow u + v = (x + w, r, -x + r - w)$

$$u + v = (x + w, r, -(x + w) + r)$$

let $x + w = f \Rightarrow u + v = (f, r, r - f)$

$$\text{so } E_1 + E_2 = \{(f, r, r - f) \in \mathbb{R}^3\}$$

or $E_1 + E_2 = \{(x, y, y - x) \in \mathbb{R}^3\}$ where $E_1 + E_2$ is also vector subspace of $\mathbb{R}^3$

Theorem:

$E_1 + E_2$ is the smallest vector subspace that contains both of E_1 and E_2 .

2) Complementary vector subspaces:

E_1 and E_2 are complementary in V denoted as $E_1 \oplus E_2 = V$

if:
$$\begin{cases} E_1 + E_2 = V \\ E_1 \cap E_2 = \{0_V\} \end{cases}$$

example:

$E_e = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(-x) = f(x)\}$ is vector subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$

$E_o = \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) \mid f(-x) = -f(x)\}$ is vector subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$

① for $f \in E_e \cap E_o \Rightarrow f \in E_e \wedge f \in E_o$

$$\Rightarrow f(-x) = f(x) \wedge f(-x) = -f(x) \Rightarrow f(x) = -f(x)$$

$$\Rightarrow f(x) = 0 \Rightarrow f = 0_{\mathcal{F}(\mathbb{R}, \mathbb{R})}$$

$$\therefore E_e \cap E_o = \{0_{\mathcal{F}(\mathbb{R}, \mathbb{R})}\}$$

② let's take two functions:

$$f_e: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \rightarrow f_e(x) = \frac{f(x) + f(-x)}{2}$$

where $f \in \mathcal{F}(\mathbb{R}, \mathbb{R})$

we have: $f_e(-x) = \frac{f(-x) + f(-(-x))}{2}$

$$f_e(-x) = \frac{f(-x) + f(x)}{2}$$

$$f_e(-x) = \frac{f(x) + f(-x)}{2} \Rightarrow f_e(-x) = f_e(x) \Rightarrow f_o(-x) = -f_o(x)$$

$$\Rightarrow f_e \in E_e$$

$$f_o: \mathbb{R} \rightarrow \mathbb{R}$$

$$x \rightarrow f_o(x) = \frac{f(x) - f(-x)}{2}$$

we have: $f_o(-x) = \frac{f(-x) - f(-(-x))}{2}$

$$f_o(-x) = \frac{f(-x) - f(x)}{2} = -\frac{f(x) - f(-x)}{2}$$

$$\Rightarrow f_o \in E_o$$

and $(f_e + f_o)(x) = f_e(x) + f_o(x) = \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2} = f(x)$

$$(f_e + f_o)(x) = f(x) \Rightarrow E_e + E_o = \mathcal{F}(\mathbb{R}, \mathbb{R})$$

$$\textcircled{1} \wedge \textcircled{2} \Rightarrow E_e \oplus E_o = \mathcal{F}(\mathbb{R}, \mathbb{R}) \quad \left(\begin{array}{l} \text{last question of semester} \\ \text{one exercise} \end{array} \right)$$